Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:

Hypertensive disorders in pregnancy

including Pre- Eclampsia, HELLP Syndrome and Acute Fatty Liver Disease

Northern Maternity Services

SUMMARY

This guideline outlines the process of Hypertension, Gestational hypertension, Pre-Eclampsia, Eclampsia, HELLP Syndrome, Acute Fatty Liver Disease.

KEY POINTS

The essential elements of this guideline are:

- Chronic hypertension
- Gestational hypertension
- Pre-eclampsia
- Severe pre-eclampsia
- Eclampsia
- HELLP syndrome
- Acute Fatty Liver disease

CONTENTS

Section		Page
1	Introduction	3
2	Background	3
3	Definitions	3
4	Responsibilities	4
5	Chronic Hypertension	5
6	Gestational Hypertension	7
7	Pre-eclampsia	10
8	Severe Pre-eclampsia	17
9	Eclampsia	22
10	HELLP Syndrome	24
11	Acute Fatty Liver Disease	25
12	Monitoring Compliance with this Guideline	27
13	References	27
14	Publication details	28
Appendices		
Appendix A	BSOTS Hypertension Triage	29
Appendix B	Risk factors for PPH	30
Appendix C	Clinical Pathway/algorithm-Roche Elecsys immunoassay sFlt-1/PLGF ratio	31
	Recommended cut-offs for the Elecsys immunoassay sFlt-1/PLGF ratio	32
Appendix D	Anti-hypertensive Therapy Tables	33
	Medicines and breastfeeding interaction table	34
Appendix E	Magnesium sulphate (MgSO4) Therapy	35
Appendix F	Letter to Community on Discharge	37

1. INTRODUCTION

1.1 The purpose of this document is to detail the process for management of hypertensive disorders of pregnancy.

1.2 Hypertensive disorders during pregnancy remain one of the leading causes of maternal death in the UK. It has potential harmful effects on both mother and baby range in severity from trivial to life threatening.

1.3 Care standards have been identified as a defining feature in determining outcome. Furthermore, hypertensive disorders can result in substantial long-term maternal morbidity including acute kidney failure, coagulopathy, chronic hypertension, cardiovascular and cerebrovascular disorders. Risks for the baby will include stillbirth, preterm delivery, Small-for-gestational-age (SFGA) and/or Intrauterine Growth Restriction (IUGR) with the latter carrying their own long-term morbidities.

1.4 The policy applies to all Trust staff involved in the care of pregnant women or women who have given birth within the previous six weeks. Implementation of this policy is up to date with recent evidence-based practice. It will ensure that all staff are aware of the clinical signs indicative of a hypertensive disorder, and how to effectively assess and treat hypertensive disorders in pregnancy.

2. BACKGROUND

2.1 This guideline reflects the NICE guideline titled “Hypertension in pregnancy: diagnosis and management” Published 25 June 2019 and Last updated: 17 April 2023

3. DEFINITIONS

3.1 Chronic Hypertension

Hypertension that is present at the booking visit or before 20 weeks or if the woman is already taking antihypertensive medication (pre-existing or persist for more than 12 weeks after pregnancy) when referred to maternity services. It can be primary or secondary in aetiology.

3.2 Gestational Hypertension

New onset hypertension presenting after 20 weeks without significant proteinuria.

3.3 Pre-Eclampsia superimposed upon existing Hypertension

New onset proteinuria.

3.4. Pre-Eclampsia

New onset hypertension presenting after 20 weeks with significant proteinuria at least 300mg/24hr. Returns to normal blood pressure within 6 weeks after delivery.

3.4.1 Severe Pre-Eclampsia

Pre-eclampsia with severe hypertension (>160/>110mmhg) that does not respond to treatment or is associated with ongoing or recurring severe headaches, visual scotomata, nausea or vomiting, epigastric pain, oliguria and severe hypertension, as well as progressive deterioration in laboratory blood tests such as rising creatinine or liver transaminases or falling platelet count, or failure of fetal growth or abnormal doppler findings.

3.5 Eclampsia

Convulsive condition associated with pre-eclampsia.

3.6 HELLP Syndrome

Haemolysis, **E**levated **L**iver enzymes and **L**ow **P**latelet count along with the diagnosis of preeclampsia.

3.7 Acute Fatty Liver Disease

Forms part of the spectrum of hypertensive disorders unique to pregnancy. It is characterised by microvesicular steatosis in the liver. The presenting symptoms are caused by acute liver insufficiency.

3.8 Significant proteinuria

Significant proteinuria is ≥30 mg/mmol in a protein: creatinine ratio (PCR) urinary sample or ≥300 mg protein in a 24-hour urine collection.

3.9 Hypertension

Mild hypertension:

diastolic blood pressure 90–99 mmHg,

systolic blood pressure 140 - 149 mmHg.

Moderate hypertension:

diastolic blood pressure 100–109 mmHg,

systolic blood pressure 150–159 mmHg.

Severe hypertension:

diastolic blood pressure 110 mmHg or greater,

systolic blood pressure 160 mmHg or greater.

4. RESPONSIBILITIES

4.1 Role of the Midwife

4.1.1 The midwife is responsible for:

- ensuring that any woman who presents with signs and symptoms of a hypertensive disorder has a complete set of observations recorded, scored on a MEOWS chart and escalated promptly,
- completing the Trust Maternity Hypertensive Disorder Screening Tool,
- ensuring that treatment is administered promptly,
- referring to the Obstetrician and multi-disciplinary team as appropriate; Medical registrar/Outreach & Resuscitation Team/ Obstetric Anaesthetist for deterioration of the seriously ill Obstetric patient, Obstetric Anaesthetist to request assessment of anaesthesia/pain relief options when a delivery is planned, Haematologist regarding treatment plan.
- comprehensively recording fluid input and output,
- ensuring any investigation results are followed-up promptly,
- ensuring on-going monitoring and observations are correctly recorded, scored and escalated as appropriate.

4.2 Role of the Obstetrician

4.2.1 The obstetrician is responsible for:

- ensuring that any woman who presents with signs and symptoms of a hypertensive disorder has a prompt triage, comprehensive and focussed assessment complete with a clearly documented plan of care,
- initiating a prompt plan of treatment in adherence with this guideline,
- liaison with the Obstetric Anaesthetist, Consultant Haematologist, Medical registrar and other multi-disciplinary team as appropriate; for example, for deterioration of the seriously ill Obstetric patient, contact Obstetric Anaesthetist regarding treatment plan assessment of anaesthesia/pain relief options when delivery planned.
- ensuring that on-going assessment, planning and treatment are undertaken frequently and documented clearly.

4.3 Role of the Obstetric Anaesthetist

4.3.1 The Obstetric Anaesthetist is responsible for:

- ensuring that any woman who is at risk of deterioration, or is already critically unwell, due to a hypertensive disorder of pregnancy has a prompt comprehensive and focussed assessment complete with a clearly documented care plan,
- liaising with the multi-disciplinary team to ensure the critically ill woman receives the appropriate level of care in the appropriate location.

5. CHRONIC HYPERTENSION

5.1 Antepartum

5.1.1 Therapeutic advice:

Women with chronic hypertension should have antihypertensive treatment dependent on pre-existing treatment, side-effect profiles and teratogenicity.

Stop ACE inhibitors or ARBs in pregnancy, preferably within 2 working days of notification of pregnancy, and offer alternatives.

Advise women who take ACE inhibitors, ARBs or thiazide/ thiazide-like diuretics that there is an increased risk of congenital abnormalities or neonatal complications if these are taken during pregnancy. Tell women who take antihypertensive treatments other than ACE inhibitors, ARBs or chlorothiazide that the limited evidence available has not shown an increased risk of congenital malformation with such treatments.

Offer pregnant women with chronic hypertension advise on:

- Weight management
- Reduced, quit smoking
- Exercise
- Healthy eating
- Lowering the amount of salt in diet

5.1.2 Therapeutic goals:

Offer antihypertensive treatment to pregnant women who have chronic hypertension and who are not already on treatment if they have:

- sustained systolic blood pressure more than or equal 140 mmHg
- sustained diastolic blood pressure more than or equal 90 mmHg

Aim for a target blood pressure of 135/85 mmHg

Consider labetalol as a first choice to treat chronic hypertension in pregnant women. Consider nifedipine for women in whom labetalol is contraindicated or not suitable (such as severe asthma or heart failure), or methyldopa if both labetalol and nifedipine are not suitable. Base the choice on any pre-existing treatment, side-effect profiles, risks (including fetal effects) and the woman's preference.

Continue with existing antihypertensive treatment if safe in pregnancy, or switch to an alternative treatment, unless:

- sustained systolic blood pressure is less than 110 mmHg or
- sustained diastolic blood pressure is less than 70 mmHg or
- the woman has symptomatic hypotension

5.1.3 Antiplatelet agents:

Offer all pregnant women with chronic hypertension aspirin 150 mg once daily starting at 12 weeks gestation until delivery, or until 36 weeks gestation in cases where there is an increased risk of primary postpartum haemorrhage (PPH.)

See Appendix B: Risk factors of PPH

5.1.4 Suspected short-term prediction of pre-eclampsia:

Offer placental growth factor (PLGF)-based testing (Roche's sFlt-1/PLGF ratio is the available test at NDDH) to help predict pre-eclampsia in the short term in pregnant women from 24 weeks to 36 weeks plus 6 days with chronic hypertension, together with other diagnostic and clinical information.

Offer PLGF based testing to help diagnose pre-eclampsia in pregnant women from 20 weeks to birth with chronic hypertension, together with other diagnostic and clinical information.

See Appendix C for Recommended cut-offs for the Elecsys immunoassay sFlt-1/PLGF ratio.

5.1.5 Fetal monitoring and antenatal appointments:

Serial fetal growth scans (3-4 weekly) from 28, 32 and 36 weeks carry out:

- ultrasound fetal growth and amniotic fluid volume assessment
- umbilical artery doppler velocimetry

Only carry out CTG if clinically indicated.

Antenatal appointments: In women with chronic hypertension, schedule additional antenatal appointments based on the individual needs of the woman and her baby. This may include:

- weekly appointments if hypertension is poorly controlled
- appointments every 2 to 4 weeks if hypertension is well-controlled.

5.2 Intrapartum

5.2.1 Timing of delivery:

Do not recommend delivery before 37 weeks gestation for women with chronic hypertension whose blood pressure is lower than 160/110 mmHg, with or without antihypertensive treatment.

For women with chronic hypertension whose blood pressure is lower than 160/110 mmHg after 37 weeks, with or without antihypertensive treatment, timing of birth, and maternal and fetal indications for birth should be agreed between the woman and the SAS Grade or Consultant Obstetrician.

Offer birth to women with refractory severe chronic hypertension, after a course of corticosteroids (if required) has been completed.

If planned early birth, offer antenatal corticosteroids and magnesium sulphate if indicated, in line with the NICE guideline on preterm labour and birth.

The delivery plan made by the Consultant Obstetrician will include a plan for maternal and fetal monitoring during labour and birth.

Attention should be paid to the frequency of blood pressure monitoring and plans to stabilise if it becomes severe or uncontrolled. Plans for fetal monitoring should be documented. Finally, haematological and biochemical monitoring should be considered in addition to plans for regional anaesthesia.

5.3 Postpartum

In women with chronic hypertension who have given birth:

- measure blood pressure:
  - daily for the first two days after birth
  - at least once between day 3 and day 5 after birth in the community
  - as clinically indicated if antihypertensive treatment is changed
- aim to keep blood pressure lower than 140/90 mmHg
- continue antenatal antihypertensive treatment/ tapering the dose accordingly
- review long-term antihypertensive treatment 2 weeks after the birth.

If a woman has taken methyldopa to treat chronic hypertension during pregnancy, stop within 2 days of birth and restart the antihypertensive treatment the woman was taking before she planned the pregnancy.

Offer women with chronic hypertension a medical review at the postnatal review (6–8 weeks after the birth) with the pre-pregnancy care team.

6. GESTATIONAL HYPERTENSION

6.1 Antepartum

Gestational hypertension is defined as new hypertension (≥140/90mmHg) after 20 weeks without features of pre-eclampsia.

Women with gestational hypertension should be assessed for risk factors for pre-eclampsia that could require additional assessment and follow-up:

- nulliparity
- age 40 years or older
- pregnancy interval of more than 10 years
- family history of pre-eclampsia
- multi-fetal pregnancy
- BMI of 35 kg/m2 or more
- gestational age at presentation
- previous history of pre-eclampsia or gestational hypertension
- Low PappA
- Placental histology confirming placental dysfunction in a previous pregnancy
- pre-existing vascular disease
- pre-existing kidney disease

6.1.1 Management of Gestational Hypertension:

Degree of hypertension	Hypertension: blood pressure (BP) of 140/ 90 to 159/109 mmHg	Severe hypertension (160/110 mmHg or higher)
Admit to hospital	Do not routinely admit to hospital	Yes (until blood pressure is below 160/110mmHg)
Anti-Hypertensive Treatment	Offer pharmacological treatment if BP remains above 140/90 mmHg Refer to Appendix D for dosage	Offer pharmacological treatment to all women Refer to Appendix D for dosage
Target blood pressure once on antihypertensive treatment	Aim for BP of 135/85 mmHg or less	Aim for BP of 135/85 mmHg or less
Measure blood pressure	Once or twice a week (depending on BP) until BP is 135/85 mmHg or less	Every 15 to 30 minutes until BP is less than 160/110 mmHg. 4-hourly while inpatient, then twice weekly after discharge.
Test for proteinuria (Use an automated reagent-strip reading device for dipstick screening for proteinuria in a secondary care setting)	Once or twice a week (with BP measurement), if proteinuria 1+ send PCR	Daily while admitted, if proteinuria 1+ send PCR
Blood tests	At presentation and then weekly: Measure full blood count, liver function, renal function, urea, coagulation screen (only if platelets <100)	At presentation and then weekly: Measure full blood count, liver function, renal function, urea, coagulation screen (only if platelets <100)
Placental growth factor (PLGF)-based testing	Carry out PLGF-based testing on 1 occasion if there is suspicion of preeclampsia	Carry out PLGF-based testing on 1 occasion if there is suspicion of preeclampsia
Fetal assessment	Offer fetal heart auscultation at every antenatal appointment Carry out ultrasound assessment of the fetus at diagnosis and, if normal, repeat every 2 to 4 weeks, if clinically indicated Carry out a cardiotocography (CTG) only if clinically indicated	Offer fetal heart auscultation at every antenatal appointment Carry out ultrasound assessment of the fetus at diagnosis and, if normal, repeat every 2 to 4 weeks, if clinically indicated Carry out a cardiotocography (CTG) only if clinically indicated

6.1.2 Therapeutic goals:

Consider labetalol to treat gestational hypertension

Consider nifedipine for women in whom labetalol is not suitable, and methyldopa if labetalol or nifedipine are not suitable. Base the choice on side-effect profiles, risk (including fetal effects) and the woman's preferences.

Outpatient BP monitoring (weekly by the community midwife)

Do not offer bed rest in hospital as a treatment for gestational hypertension.

6.1.3 Suspected short-term prediction of pre-eclampsia:

Placental growth factor (PLGF) Test available at NDDH is Roche's sFlt-1/PLGF ratio:

Offer (PLGF)-based testing to help predict pre-eclampsia in the short term in pregnant women from 24 weeks to 36 weeks plus 6 days, together with other diagnostic and clinical information.

Offer PLGF based testing to help diagnose pre-eclampsia in pregnant women from 20 weeks to birth, together with other diagnostic and clinical information.

(See Appendix C for Recommended cut-offs for the Elecsys immunoassay sFlt-1/PLGF ratio.)

6.1.4 Fetal monitoring and antenatal appointments:

Carry out an ultrasound for fetal growth and amniotic fluid volume assessment and umbilical artery doppler velocimetry at diagnosis and if normal repeat every 2 to 4 weeks, if clinically indicated.

Only carry out CTG if clinically indicated.

6.1.5 Timing of delivery:

Do not offer birth before 37 weeks to women with gestational hypertension whose blood pressure is lower than 160/110 mmHg, with or without antihypertensive treatment.

For women with gestational hypertension whose blood pressure is lower than 160/110 mmHg after 37 weeks, with or without antihypertensive treatment, timing of birth, and maternal and fetal indications for birth should be agreed between the woman and the SAS Grade or Consultant Obstetrician.

The delivery plan made by the Consultant Obstetrician will include a plan for maternal and fetal monitoring during labour and birth.

If planned early birth is necessary, offer a course of antenatal corticosteroids (after discussion with the patient) and magnesium sulphate if indicated (from 24 weeks to 33+6 weeks), in line with the NICE guideline on preterm labour and birth.

6.2 Intrapartum

Attention should be paid to the frequency of blood pressure monitoring and plans to stabilise if it becomes severe or uncontrolled. Plans for fetal monitoring should be documented (needs continuous electronic fetal monitoring CEFM in labour). Finally, haematological and biochemical monitoring should be considered in addition to plans for regional anaesthesia.

6.3 Postpartum

In women with gestational hypertension who have given birth, measure blood pressure:

- daily for the first 2 days after birth
- at least once between day 3 and day 5 after birth in the community.
- as clinically indicated if antihypertensive treatment is changed after birth

In women with gestational hypertension who have given birth:

- continue use of antenatal antihypertensive treatment
- reduce antihypertensive treatment if their blood pressure falls below 130/80 mmHg.

If a woman has taken methyldopa to treat gestational hypertension, stop within 2 days of birth.

For women with gestational hypertension who did not take antihypertensive treatment and have given birth, start antihypertensive treatment if their blood pressure is higher than 149/99 mmHg.

Ensure there is a clearly documented Obstetric care plan for women with gestational hypertension who have given birth and are being transferred to community care that includes all of the following:

- who will provide follow-up care, including medical review if needed
- frequency of blood pressure monitoring needed
- thresholds for reducing or stopping treatment
- indications for referral to primary care for blood pressure review

Refer for medical review all women who have had gestational hypertension and remain on antihypertensive treatment 2 weeks after transfer to the community.

Ensure that all women who have had gestational hypertension have an appointment for a medical review at the postnatal review (6–8 weeks post birth). If antihypertensive treatment is still required following assessment at the postnatal review, specialist assessment should be arranged.

7. PRE-ECLAMPSIA

The maternal death rate from preeclampsia and eclampsia remains more than five times higher than the lowest observed rate in 2012-14 (MBBRACE, 2021).

Pre-eclampsia is defined as new onset of hypertension after 20 weeks with one or more of the following:

- Proteinuria (PCR ≥ 30)
- Other maternal organ dysfunction:
  - **Renal insufficiency** (creatinine 90 micromol/litre or more, 1.02 mg/100 ml or more).
  - **Liver involvement** (elevated transaminases [alanine aminotransferase or aspartate aminotransferase over 40 IU/litre] with or without right upper quadrant or epigastric abdominal pain).
  - **Neurological complications** such as eclampsia, altered mental status, blindness, stroke, clonus, severe headaches or persistent visual scotomata.
  - **Haematological complications** such as thrombocytopenia (platelet count below 150,000/microlitre), disseminated intravascular coagulation or haemolysis
  - **Uteroplacental dysfunction** such as fetal growth restriction, abnormal umbilical artery doppler waveform analysis, or stillbirth.

7.1 Antepartum

7.1.1 Antiplatelet agents:

Advise pregnant women who have 1 high risk factor/more than 1 moderate risk factor of pre-eclampsia to take aspirin 150 mg daily starting at 12 weeks gestation until delivery, or until 36 weeks gestation in cases where there is an increased risk of bleeding. See Appendix B: Risk factors of PPH.

(If under 16 years of age, decision to start aspirin should be made by a consultant.)

7.1.2 Risk factors of developing pre-eclampsia:

High risk factors:

hypertensive disease during a previous pregnancy
chronic kidney disease
autoimmune disease such as systemic lupus erythematosus or
antiphospholipid syndrome
type 1 or type 2 diabetes
chronic hypertension.

Moderate risk factors:

nulliparity
age 40 years or older
pregnancy interval of more than 10 years
body mass index (BMI) of 35 kg/m2 or more at first visit
family history of pre-eclampsia
multi-fetal pregnancy (twin pregnancy)
Low PappA
Placental histology confirming placental dysfunction in a previous pregnancy

Symptoms of pre-eclampsia:

severe headache
problems with vision, such as blurring or flashing before the eyes
severe pain just below the ribs
vomiting
sudden swelling of the face, hands or feet

7.1.3 Suspected short-term prediction of pre-eclampsia:

- Placental growth factor (PLGF) Test available at NDDH is Roche's sFlt-1/PLGF ratio:
- Offer (PLGF)-based testing **help predict pre-eclampsia** in the short term in pregnant women from **24 weeks to 36 weeks plus 6 days**, together with other diagnostic and clinical information.
- Offer PLGF based testing to **help diagnose pre-eclampsia** in pregnant women from **20 weeks to birth**, together with other diagnostic and clinical information.

(See Appendix C for Recommended cut-offs for the Elecsys immunoassay sFlt-1/PLGF ratio.)

- Do not take in multiple fetal pregnancies.
- Do not use PLGF-based tests to make decisions about whether to offer a planned early birth to people with preterm pre-eclampsia.
- Use a PLGF-based test once per episode of suspected preterm pre-eclampsia. NICE has recommended testing just once when a person presents with possible symptoms of preterm pre-eclampsia (an episode).

7.1.4 Management of Pre-Eclampsia:

Degree of hypertension	Hypertension: blood pressure (BP) of 140/90 to 159/109 mmHg	Severe hypertension (160/110 mmHg or higher)
Admit to hospital	Admit if any clinical concerns for the wellbeing of the woman or baby or if high risk of adverse events	Admit, but if BP falls below 160/110 mmHg, then manage as for hypertension
Anti-Hypertensive Treatment	Offer pharmacological treatment if BP remains above 140/90 mmHg. Aim for BP of 135/85 mmHg or less Refer to Appendix D for dosage	Offer pharmacological treatment to all women Aim for BP of 135/85 mmHg or less Refer to Appendix D for dosage
Measure blood pressure	Every 4 hours if the woman is admitted to hospital	Every 15 to 30 minutes until BP is less than 160/ 110 mmHg, then at least 4 times daily while the woman is an inpatient, depending on clinical circumstances
Test for proteinuria	Only repeat if clinically indicated, for example, if new symptoms and signs develop or if there is uncertainty over diagnosis	Only repeat if clinically indicated, for example, if new symptoms and signs develop or if there is uncertainty over diagnosis
Blood tests	At presentation and then weekly: Measure full blood count, liver function, renal function, urea, coagulation screen (only if platelets <100) earlier if indicated	At presentation and then weekly: Measure full blood count, liver function, renal function, urea, coagulation screen (only if platelets <100) earlier if indicated
Fetal Assessment	Offer fetal heart auscultation at every antenatal appointment Carry out ultrasound fetal growth every 2-4 weeks from 26/40 until delivery Carry out a cardiotocography (CTG) at diagnosis and then only if clinically indicated	Offer fetal heart auscultation at every antenatal appointment Carry out ultrasound fetal growth every 2-4 weeks from 26/40 until delivery Carry out a CTG at diagnosis and then only if clinically indicated

7.1.5 Therapeutic Goals:

Women with pre-eclampsia must have a comprehensive assessment performed by a healthcare professional trained in the management of hypertensive disorders of pregnancy at each consultation.

Women who present in the antenatal period with signs and symptoms of pre- eclampsia must have a comprehensive assessment using the NDDH tool in Triage including blood pressure, testing for proteinuria and blood tests. Clinical diagnosis should be made using the definitions at the beginning of this guideline and a clear further plan documented.

7.1.5.1 The treatment regimen will focus on:

- controlling the blood pressure to reduce the risk of mortality and morbidity associated with hypertension such as stroke and reducing the risk of seizure
- offer labetalol to treat hypertension in pregnant women with preeclampsia. Offer nifedipine for women in whom labetalol is contraindicated/not suitable, and methyldopa if labetalol or nifedipine are not suitable. Base the choice on any pre-existing treatment, side-effect profiles, risks (including fetal effects) and the woman's preference.

Refer to Appendix D for dosage

7.1.6 Fetal monitoring (pre-eclampsia or severe gestational hypertension):

- Carry out CTG at diagnosis
- If conservative treatment is planned, carry out all the following tests at diagnosis:
  - ultrasound for fetal growth and amniotic fluid volume assessment
  - umbilical artery doppler velocimetry
- If the results of all fetal monitoring are normal, do not routinely repeat CTG unless clinically indicated
- Repeat CTG if any of the following occur:
  - the woman reports a change in fetal movement
  - vaginal bleeding
  - abdominal pain
  - deterioration in maternal condition
- Repeat ultrasound for fetal growth and amniotic fluid volume assessment or umbilical artery doppler velocimetry every 2 weeks, with subsequent surveillance and monitoring determined by the findings of these scans.

7.1.6.1 Women who need additional Fetal monitoring:

Carry out an ultrasound for fetal growth and amniotic fluid volume assessment and umbilical artery doppler velocimetry starting at between 28 and 30 weeks (or at least 2 weeks before previous gestational age of onset if earlier than 28 weeks) and repeating 4 weeks later in women with previous:

- severe pre-eclampsia
- pre-eclampsia that resulted in birth before 34 weeks
- pre-eclampsia with a baby whose birth weight was less than the 10th centile
- intrauterine death
- placental abruption

7.2 Intrapartum

7.2.1 Planning delivery:

Women who have pre-eclampsia with hypertension may progress rapidly to severe disease with its associated risks, however until severe disease manifests it may be unclear whether the risks of pre-eclampsia outweigh the risks of planned late preterm birth for the baby.

Management of the timing of delivery must therefore be decided by the Consultant Obstetrician with a clearly documented plan in the woman’s notes including the maternal (biochemical, haematological and clinical) and fetal thresholds for delivery.

7.2.2 Timing of delivery:

Conservative management until 34 weeks should be the first line response where the blood pressure is well controlled with an effective treatment regimen and there is a clearly followed monitoring plan.

7.2.2.1 Elective delivery will be offered to women with pre-eclampsia before 34 weeks gestation if:

- severe hypertension develops refractory to treatment
- maternal or fetal indications develop as specified in the consultant plan
- discussion with neonatal and anaesthetic teams are complete
- offer IV Magnesium Sulphate and a course of corticosteroids in line with NICE guideline on preterm labour and birth

7.2.2.2 Elective delivery will be offered to women with pre-eclampsia at 34 to 36+6 weeks gestation if:

- severe hypertension is present and stabilised,
- maternal or fetal indications develop as specified in the consultant plan,
- after discussion with neonatal and anaesthetic teams
- offer a course of corticosteroids in line with NICE guideline on preterm labour and birth

7.2.2.3 Elective delivery can be offered to women with pre-eclampsia after 37 weeks gestation, within 24–48 hours of a new diagnosis.

Emergency delivery should be recommended at any gestation where the risks of the disease outweigh the benefits of the fetus remaining in utero. The discussion for emergency delivery should be undertaken with the woman and her family, the Consultant Obstetrician and Consultant Paediatrician.

The delivery plan made by the Consultant Obstetrician will include a plan for maternal and fetal monitoring during labour and birth/prior delivery.

Attention should be paid to the frequency of blood pressure monitoring and plans to stabilise if it becomes severe or uncontrolled. Plans for fetal monitoring should be documented. Finally, haematological and biochemical monitoring should be considered in addition to plans for regional anaesthesia.

7.2.3 Fetal monitoring

The delivery plan made by the Consultant Obstetrician will include a plan for fetal monitoring during labour and birth.

Continuous electronic fetal monitoring (CEFM) for labour and birth is recommended for all women with hypertensive disorders. Attention should be paid to:

- the most recent estimated fetal weight on scan in addition to the symphysio-fundal height measurement made by the midwife on abdominal palpation.
- the features of the assessment of the fetal heart which may be affected by the anti-hypertensive medication and placental insufficiency.

7.2.4 Maternal monitoring and assessment.

- Continue use of antenatal antihypertensive treatment during labour.
- **Blood Pressure** monitoring will be determined by the severity of the hypertension. Women with mild or moderate hypertension that is well controlled should have hourly blood pressure monitoring. This should be increased if the findings become severe with a documented review and plan by a SAS Grade or Consultant Obstetrician until the blood pressure is within acceptable ranges.

- Women with severe hypertension should have continuous monitoring with a documented review and plan by a SAS Grade or Consultant Obstetrician until the blood pressure is within acceptable ranges. During labour, measure BP every 15-30 minutes until BP <160/110 mmHg in women with severe hypertension.
- **Haematological and biochemical monitoring** should be considered on an individual basis determined by the gestation, mode and length of labour (e.g. IOL pre-term for Pre-eclampsia vs. spontaneous labour at term), the recent haematological and biochemical markers and the severity of symptoms.
- Analgesia and anaesthesia options should be made available to all women in labour as per Trust guidance. Specific considerations for women with hypertensive disorders will include the haematological and biochemical markers, primarily platelet count, and the withholding of the standard 1 litre pre-load for epidural anaesthesia for women with severe pre-eclampsia.
- Strict **fluid balance** monitoring should be completed throughout labour for women with severe hypertension.

7.2.5 Second stage:

- The duration of second stage of labour should not be routinely limited for all women with hypertensive disorders. An assessment of individual risk factors should be documented along with a plan that has been discussed and agreed with the woman.
- Women with well controlled mild or moderate hypertension should follow routine guidance for management of the second stage as per Intrapartum guidelines.
- Women with severe hypertension or unstable hypertension should have regular assessments during the second stage paying attention to descent of the presenting part and signs of progress in the second stage. If there is no demonstrable progress or descent of the presenting part, and the blood pressure is worsening, assisted birth including operative delivery may be required and an Obstetric review should be completed with a view to expediting delivery.

7.2.6 Third stage:

- Avoid ergometrine when administering uterotonics for active management of the third stage.

7.3 Postpartum

7.3.1 In women with pre-eclampsia who did not take antihypertensive treatment and have given birth, start antihypertensive treatment if blood pressure is 150/100 mmHg or higher

Measure blood pressure:

- at least four times a day while the woman is an inpatient,
- at least once between day 3 and day 5 after birth,
- on alternate days until normal if blood pressure was abnormal on days 3–5.

7.3.2 In women with pre-eclampsia who have taken antihypertensive treatment and have given birth, continue antenatal antihypertensive treatment. Consider reducing antihypertensive treatment if their blood pressure falls below 140/90 mmHg and reduce antihypertensive treatment if their blood pressure falls below 130/80 mmHg.

7.3.3 If a woman has taken methyldopa to treat hypertension, stop within 2 days of birth.

Measure blood pressure:

- at least four times a day while the woman is an inpatient
- every 1–2 days for up to 2 weeks after transfer to community care until the woman is off treatment and has no hypertension.

Ask women with pre-eclampsia who have given birth about symptoms of severe headache, epigastric pain and pre-eclampsia symptoms each time the blood pressure is measured.

Fluid balance monitoring can cease when the hypertension is well controlled and creatinine levels are within the normal range.

In PPH, if on fluid restriction, consider 40 units oxytocin in 100 mls and deliver at 25mls per hour over 4 hours

Thromboprophylaxis, assess risk and consider LMWH as per protocol

Haematological and biochemical monitoring should be completed no less than 48–72 hours after birth or step-down from critical care for women with mild or moderate hypertension. Do not repeat if the results are all completely normal.

If the haematological and biochemical markers are improving but stay within the abnormal range, repeat again within 48 hours.

If the haematological and biochemical markers are not improving, repeat again within 24hours and ensure a fully documented Obstetric assessment and plan is completed.

Women with pre-eclampsia who have given birth can be discharged home and transferred to community care if all of the following criteria have been met:

- there are no symptoms of pre-eclampsia
- blood pressure, with or without treatment, is 149/99 mmHg or lower
- blood test results are stable or improving.

Ensure there is a written care plan that includes all of the following:

- who will provide follow-up care, including medical review if needed
- frequency of blood pressure monitoring
- thresholds for reducing or stopping treatment
- indications for referral to primary care for blood pressure review
- self-monitoring for symptoms.

Postnatal follow-up:

- 2 weeks after transfer to community care a medical review must be completed for women who are still on antihypertensive treatment.
- 6– 8 weeks after the birth a medical review must be completed for all women who have had pre-eclampsia.
- 6–8 weeks after the birth the postnatal review must include referral for a specialist assessment for all women who have had pre-eclampsia and who still need antihypertensive treatment at the postnatal review.
- Advice and document use of aspirin 150 mg in next pregnancy from 12 weeks onward.

In women with pre-eclampsia who have given birth, carry out a urinary reagent-strip test at the postnatal review (6–8 weeks after the birth).

Offer women who had pre-eclampsia and still have proteinuria (1+ or more) at the postnatal review (6–8 weeks after the birth) a further review at 3 months after the birth to assess kidney function and consider offering them a referral for specialist kidney assessment.

8. SEVERE PRE-ECLAMPSIA

Severe pre-eclampsia continues to cause maternal and perinatal morbidity. The standard of care women receive in managing this critical illness has a defining impact for the mother and the baby. The primary defining feature in effective management is to make a clear diagnosis as this will set in motion a definite plan of management.

8.1 Diagnosis of Severe Pre-Eclampsia based on the NICE (2023) definition:

'Pre-eclampsia with severe hypertension that does not respond to treatment or is associated with ongoing or recurring severe headaches, visual scotomata, nausea or vomiting, epigastric pain, oliguria and severe hypertension, as well as progressive deterioration in laboratory blood tests such as rising creatinine or liver transaminases or falling platelet count, or failure of fetal growth or abnormal doppler findings.'

In practice, you will see:

- severe hypertension
- <u>and</u> pre-eclampsia

In addition, you <u>*may also*</u> see:

- symptoms of severe headache, oedema, visual disturbances, epigastric pain, vomiting, papilledema, signs of clonus (≥3 beats), liver tenderness.
- abnormal bloods including:
  - abnormal liver enzymes (ALT or AST rising to above 50 IU/litre,
  - platelet count falling below 150
  - LDH>525
  - renal insufficiency (creatinine 90 micromol/litre or more, 1.02 mg/100 ml or more)

8.2 Antepartum

Diagnosis of severe pre-eclampsia will come with an assessment of, and discussion about, the need for delivery. This will be undertaken with the woman and guidance from the Paediatric team about location and timing of delivery.

Acute management of severe pre-eclampsia will be conducted in the inpatient setting appropriate to the woman's needs. This may be on delivery suite prior to delivery with attendance by the wider multi-disciplinary team as appropriate or it may be in the high dependency or ICU setting. It may also require transfer out for maternal or fetal reasons and this option should be readily and promptly considered in the best interests of the mother and the baby.

8.2.1 <u>Location of care</u>

Women with severe pre-eclampsia should be referred to the appropriate expertise within the multi-disciplinary team. This will include the Obstetric Anaesthetist and may further include the ICU Anaesthetist and Outreach Resuscitation team. The location of care will be determined by the multi-disciplinary team and the decision will be based on the location where the needs of the woman can be best met. In general terms, this will be as follows;

Level 3 care

(ICU) for women with severe pre-eclampsia with any of the following complications:

- needing ventilation
- evidence of cardiac failure

Level 2 care

(ICU/HDU/CDS) for women with severe pre-eclampsia with any of the following complications:

- eclampsia
- HELLP syndrome
- haemorrhage
- hyperkalaemia
- severe oliguria
- coagulation support
- intravenous antihypertensive treatment
- initial stabilisation of severe hypertension
- abnormal neurology

Level 1 care

- Ongoing conservative antenatal management of severe preterm hypertension
- Step-down treatment after the birth

8.2 Intrapartum

8.2.1 Therapeutic goals

The goals in management of severe pre-eclampsia are:

- control hypertension
- reduce risk of seizure
- precise fluid restriction and fluid balance
- plan to deliver

8.2.2 Reducing the risk of seizure in Severe Pre-Eclampsia

Once severe pre-eclampsia is diagnosed commence intravenous magnesium sulphate.

Administration of magnesium sulphate:

Loading dose: 4g $MgSO_4$ as a SLOW BOLUS over 20 minutes

- Draw up 20 mls of 20% $MgSO_4$ solution (4g) into a 50ml syringe
- Place the syringe in a syringe driver and run it at 60ml/hr
- The i.v. infusion will then run over 20 minutes
- In an emergency situation it can also be given as a slow IV push over at least 5 minutes

Maintenance Dose – Magnesium Sulphate **1 gram per Hour**

- Draw up 50ml of 20% $MgSO_4$ (10 g) in a 50 ml syringe
- Give IV using syringe driver at rate of 5ml/hour (1g/hr)

Recurrent seizures should be treated with a further dose of 2g given over 5 minutes.

Monitor:

- patellar reflexes hourly (or elbow with epidural)
- respiratory rate (>10 per min) every 15 mins
- O2 Sats >95% every 15 mins
- pulse rate every 15 mins
- urine output hourly
- MgS04 levels if toxicity suspected or persistent oliguria or known impaired renal function

See Appendix F for serum levels reference range

8.2.3 Management of severe pre-eclampsia:

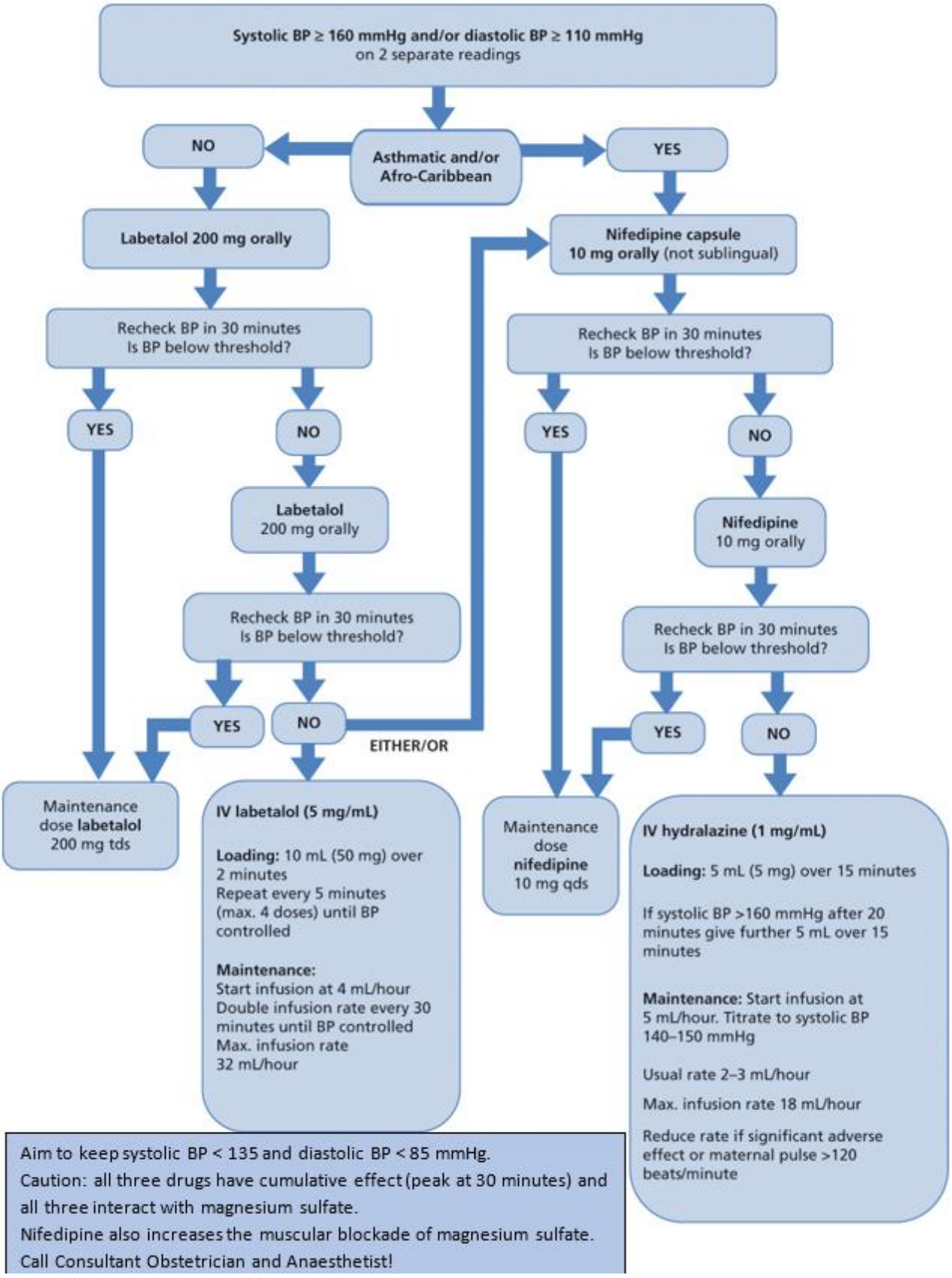
Aim: keep blood pressure <135 mmHg systolic and <85 mmHg diastolic.

Treat severe hypertension immediately with the following:

labetalol (oral or intravenous), hydralazine (intravenous), nifedipine (oral).

Refer to Appendix D for treatment regimen.

8.2.3.1 Algorithm for the management of severe hypertension (adapted from PROMPT – following NICE 2023)



8.2.4 Fluid Restriction:

In women with severe pre-eclampsia, limit total fluid input to 80 ml/hour.

Maintain a meticulous fluid balance of input and output to reduce the risk of fluid overload and pulmonary hypertension.

Urine output should be >0.5ml/kg/hr or approximately >100mls/4hrs. If there is any concern about urine output Obstetric and Anaesthetic assessment is required.

Do not use volume expansion in women with severe pre-eclampsia unless there are other on-going fluid losses (for example, haemorrhage). It may be considered as a once-off 500ml administration if hydralazine is the antenatal antihypertensive.

8.2.5 Plan for delivery:

The mode of birth and/or individualised induction of labour plan will be determined by the clinical circumstances and the woman's preference.

Management of the timing of delivery will be decided by the Consultant Obstetrician, in discussion with the woman, with a clearly documented plan in the woman's notes including the maternal (biochemical, haematological and clinical) and fetal thresholds for delivery.

Refer to Planning delivery/Timing of delivery for detailed advice per gestation.

8.2.6 Administration of antepartum corticosteroids for fetal lung maturation:

If birth is considered likely within 7 days: (after discussion with patient refer new RCOG guideline)

- offer two doses of betamethasone 12 mg intramuscularly 24 hours apart in women between 24+0 and 33+6 weeks
- consider giving two doses of betamethasone 12 mg intramuscularly 24 hours apart in women between 34+0 and 35+6 weeks.

Consider a single repeat course of maternal corticosteroids for women less than 34+0 weeks of pregnancy who:

- have already had a course of corticosteroids when this was more than 7 days ago, and
- are at very high risk of giving birth in the next 48 hours.

Do not give more than 2 courses of maternal corticosteroids for preterm birth.

8.2.7 Intrapartum maternal and fetal monitoring:

- Refer to "Intrapartum maternal and fetal monitoring of pre-eclampsia"

8.2.8 Management of the third stage:

- Avoid ergometrine when administering uterotonics for active management of the third stage.

8.3 Postpartum

Management of the woman with newly diagnosed severe pre-eclampsia in the postpartum period will rely on the therapeutic goals and management plan set out previously.

Women who were diagnosed with severe pre-eclampsia antepartum or intrapartum, and are now delivered, should follow an accurate plan of management as defined in "Postpartum pre-eclampsia"

Despite delivery, women who have pre-eclampsia may progress rapidly to severe disease with its associated risks at any stage in the postpartum period.

Ask women with pre-eclampsia who have given birth about severe headache, epigastric pain and pre-eclampsia symptoms each time the blood pressure is measured.

9. ECLAMPSIA

9.1 Eclampsia is a convulsive condition associated with pre-eclampsia. It is an obstetric emergency characterised by seizures which may be profound or subtle in nature therefore the clinician should be both vigilant and prepared for eclampsia when any hypertensive disorder of pregnancy is diagnosed.

Therapeutic goals and Management plan (Eclampsia).

Eclampsia is an Obstetric emergency that warrants an immediate call for help and emergency response as follows:

1. Call for help; you will need a senior midwife, SAS Grade or Consultant Obstetrician and Obstetric Anaesthetist. Out of hours the Consultant Obstetrician and Anaesthetist should be called to attend.
2. Assess Airway, Breathing and Circulation
3. Control seizure/s:

Loading dose: 4g $MgSO_4$ as a SLOW BOLUS over 20 minutes

- Draw up 20 mls of 20% $MgSO_4$ solution (4g) into a 50ml syringe
- Place the syringe in a syringe driver and run it at 60ml/hr
- The i.v. infusion will then run over 20 minutes
- In an emergency situation it can also be given as a slow IV push over at least 5 minutes

Maintenance Dose – Magnesium Sulphate **1 gram per Hour**

- Draw up 50ml of 20% $MgSO_4$ (10 g) in a 50 ml syringe
- Give IV using syringe driver at rate of 5ml/hour (1g/hr)

Do not use diazepam, phenytoin to magnesium sulphate in women with eclampsia.

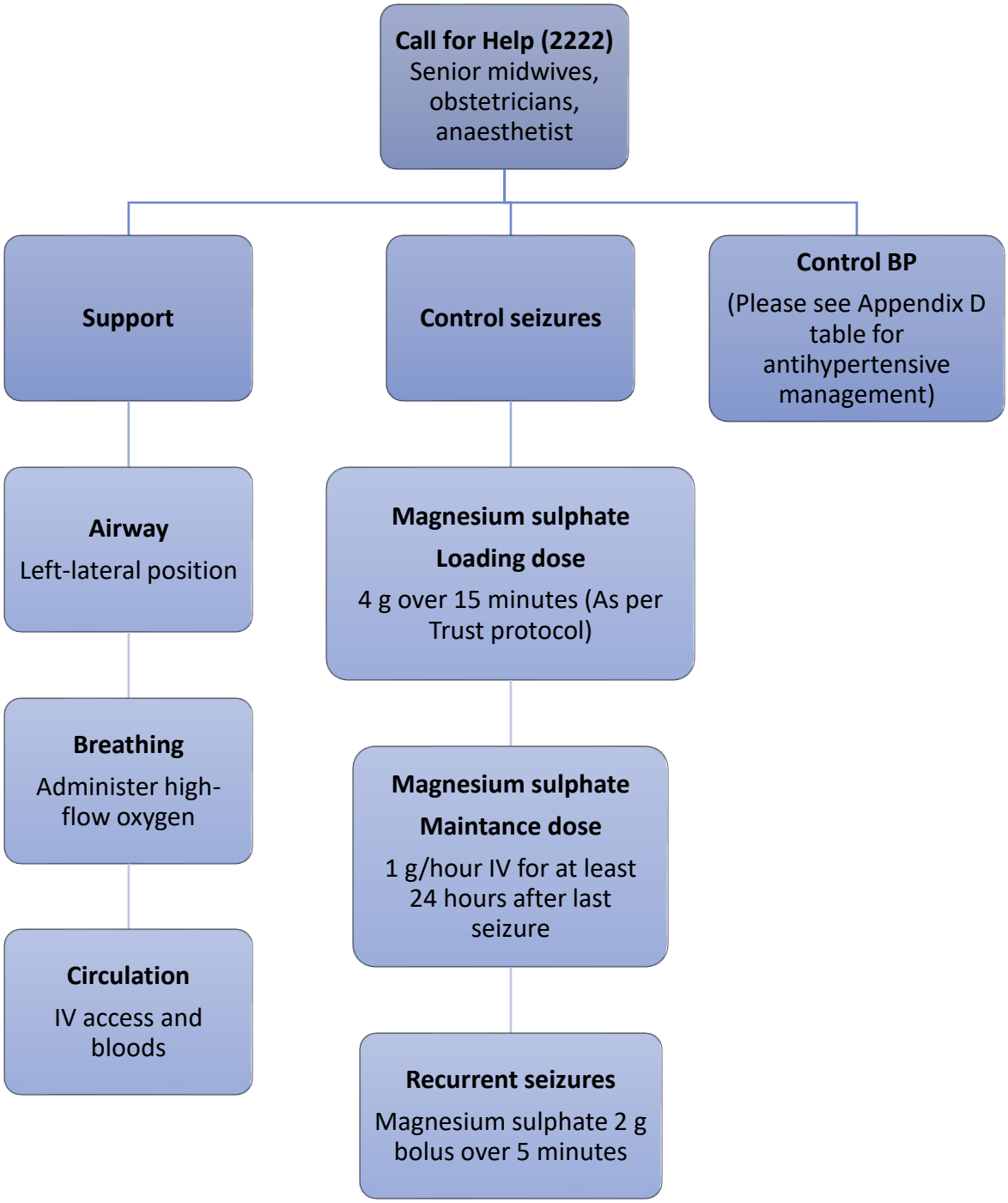
During this emergent phase, observations will need to be completed every 5mins. Ensure that:

- hypertension is managed,
- strict fluid balance is maintained,
- observations are entered in EPIC,
- make a plan for delivery (stabilise the woman's condition prior to delivery),
- a clear Obstetric plan is documented,
- a clear Anaesthetic plan is documented.

9.2

Management of Eclampsia Algorithm

PROMPT



10. HELLP SYNDROME

10.1 HELLP syndrome is a variant of severe pre-eclampsia where hypertension is less marked but where there is severe involvement of both the liver and the coagulation system. In addition to the usual complications of severe pre-eclampsia there is a risk of haemorrhage due to disseminated intravascular coagulation (DIC), liver failure and/or rupture, renal failure, eclampsia, pulmonary oedema and adult respiratory distress syndrome (RCOG, 2011).

10.2 HELLP syndrome is a life-threatening condition that may present unheralded without signs of pre-eclampsia; it can easily be confused with 'severe heartburn', non-specific illnesses or gastroenteritis. Repeated administration of analgesia for epigastric pain must not be undertaken without a full biochemical and haematological profile.

The primary defining feature in effective management is to make a prompt and clear diagnosis as this will set in motion a definite plan of management.

In practice, you may see:

- epigastric pain (the commonest symptom),
- hepatic (liver) tenderness,
- less often headaches, vomiting, malaise,
- hypertension may be mild or even absent,
- raised capillary blood glucose in the absence of diabetes.

In practice, you will see:

- deranged biochemical and haematological markers including;
- platelet count < 100x10 9/L (and/or falling from previous results)
- plasma ALT> 70 umol/L
- Bilirubin in the MSU (due to haemolysis)

10.3 Multidisciplinary team involvement and location of care

Early referral is recommended for women with HELLP syndrome to the appropriate expertise within the multi-disciplinary team. This will include the Obstetric Anaesthetist and may further include the ICU Anaesthetist, Consultant Haematologist and Outreach Resuscitation team.

The location of care will be determined by the multi-disciplinary team and the decision will be based on the location where the needs of the woman can be best met. Transfer out to a specialist unit for maternal or fetal reasons may also be required and this option should be readily and promptly considered in the best interests of the mother and the baby.

10.4 Therapeutic goals and management plan (HELLP Syndrome)

Do not use dexamethasone or betamethasone for the treatment of HELLP syndrome.

The therapeutic goals in the management of HELLP syndrome are to achieve a return to normal biochemical and haematological profiles without significant long-term complications. The management will be based on frequent and comprehensive senior multidisciplinary review of maternal stability and fetal wellbeing.

To achieve this, complete the following:

- expedite delivery,
- limit total fluid input to 80 ml/hour,
- maintain a meticulous fluid balance of input and output to reduce the risk of fluid overload and pulmonary hypertension.

- reduce the risk of seizure and manage hypertension (if it exists) as per 5.9 'Control Hypertension' and 'Reduce risk of seizure'.
- the need for platelet transfusion or fresh frozen plasma (FFP) should be considered in severe cases where the platelet count falls below 50.
- minimum thrice daily documented assessment and plan by the Consultant Obstetrician and Obstetric Anaesthetist.
- daily senior multidisciplinary liaison between the Consultant Obstetrician, Consultant Anaesthetist (covering Obstetrics), Consultant Haematologist, Consultant Paediatrician and Outreach Resuscitation team.

11. ACUTE FATTY LIVER DISEASE

11.1 Acute fatty liver disease (AFLD) is a rare but potentially fatal condition for both mother and baby with an indigence of 1 per 20,000 pregnancies. The primary complication of AFLD is that diagnosis is frequently delayed thereby resulting in a manifest condition that is difficult to reverse.

It occurs predominantly in the third trimester and forms part of a spectrum of hypertensive disorders related to pre-eclampsia and therefore carries with it the risk of the usual complications of severe pre-eclampsia in addition to multiple organ failure, haemorrhage, liver rupture, eclampsia, pulmonary oedema and adult respiratory distress syndrome.

Women at greatest risk are those who are older, primiparous, low BMI, multiple pregnancies. Twin gestation carries 14-fold increased risk for AFLD.

The primary defining feature in effective management is to make a prompt and clear diagnosis as this will set in motion a definite plan of management.

In practice, you may see;

- Renal failure,
- Hypoglycaemia,
- Infection,
- Gastrointestinal haemorrhage,
- Coagulopathy,
- Severe postpartum haemorrhage,
- Stillbirth

11.2 Acute Fatty Liver Disease is closely related to, and can be difficult to differentiate from, HELLP syndrome.

Symptom	HELLP	AFLD
Elevated Transaminases	**+**	**++**
Hypoglycaemia	**+/-**	**++**
DIC	**+**	**++**
Leucocytosis	**+**	**++**
Multiple Pregnancy	**+**	**++**
USS/CT	Normal	Bright Liver/ Ascites
Male Fetus	50%	70% (M:F = 3:1)
Hypertension	**++**	**+**
Proteinuria	**++**	**+**
Primiparous	**++**	**+**
Epigastric Pain	**+**	**+**
Thrombocytopaenia	**++**	**+/-**

11.3 Since a widely accepted diagnostic criterion does not exist for AFLD, UKOSS recommends using the criteria referred to as the Swansea Criteria.

Swansea diagnostic criteria for AFLD: 6 or more of the following in the absence of another explanation.	
NB Listed in order of clinical significance **BOLD** indicates 98-100% association with AFLD	
Clinical features	Vomiting Abdominal Pain Polydipsia/Polyuria Encephalopathy
Biochemical and/or Haematological profiles	**Bilirubin >14 umol** **AST/ALT >42 IU/l** **Haematological Leucocytosis >11 x109 /l** Endocrine Glucose < 4 mmol/l Creatinine >150 umol/l Ammonia >47 umol/l Coagulopathy- PT >14 secs OR APTT > 34 secs (often with Plt count >100 x1012)
Radiological findings	Abdominal USS showing Bright Liver echo texture/Ascites.
Histological findings.	Liver Biopsy showing microvesicular steatosis

11.4 Multidisciplinary team involvement and location of care

Early referral is recommended for women with Acute Fatty Liver Disease to the appropriate expertise within the mutli-disciplinary team. This will include the Obstetric Anaesthetist and may further include the ICU Anaesthetist, Consultant Haematologist and Outreach Resuscitation team.

The location of care will be determined by the multi-disciplinary team and the decision will be based on the location where the needs of the woman can be best met. Transfer out to a specialist unit should be readily and promptly considered in the best interests of the mother and the baby.

11.5 Therapeutic goals and Management plan (Acute Fatty Liver Disease)

The therapeutic goals in the management of Acute Fatty Liver Disease are to achieve a return to normal biochemical and haematological profiles without significant long-term complications. The management will be based on frequent and comprehensive senior multidisciplinary review of maternal stability and fetal wellbeing.

Early diagnosis of Acute Fatty Liver Disease and prompt delivery is the defining feature of effective treatment.

To achieve this, complete the following;

- Referral to a specialist Liver Unit is recommended. Transfer to NDDH ICU prior to external transfer may be considered, if CDS is the interim location of care documented review and plan by the ICU Anaesthetist is required.
- Expedite delivery: mode of delivery will most usually by LSCS with GA due to the severity of the condition and associated risk of spinal haematoma with regional anaesthesia.
- Aggressive management of coagulopathy with FFP, Crypoprecipitate, Novoseven in discussion with haematologist

- Aggressive management of hypoglycaemia with 50% intravenous glucose.
- Consider N-acetyl Cysteine as it can improve haemodynamics whilst preventing progressive decompensation.
- Have a low threshold for parenteral antibiotics given the risk of infection.
- Reduce the risk of cerebral oedema by: raising the head of the bed, use of oral Lactulose and if appropriate use of IV sedation and hyperventilation.
- Limit total fluid input to 80 ml/hour.
- Maintain a meticulous fluid balance of input and output to reduce the risk of fluid overload and pulmonary hypertension.
- Reduce the risk of seizure and manage hypertension (if it exists) as per 5.9 'Control Hypertension' and 'Reduce risk of seizure'.
- Minimum thrice daily documented assessment and plan by the Consultant Obstetrician and Obstetric Anaesthetist.
- Daily senior multidisciplinary liaison between the Consultant Obstetrician, Consultant Anaesthetist (covering Obstetrics), Consultant Haematologist, Consultant Paediatrician and Outreach Resuscitation team. Monitoring Compliance with and the Effectiveness of the Guideline

12. MONITORING COMPLIANCE WITH THIS GUIDELINE

An up to date copy of this guideline is available to all staff on the Trust HUB.

As a matter of routine, this guideline will be reviewed triennially by the Maternity Services Guideline group.

Reporting for non-compliance and review of effectiveness of the guideline will be identified through the risk process within maternity and led by appointed maternity Risk leads. The maternity services audit process will include review of this guideline. All versions of these guidelines will be archived in electronic format by the author within the Maternity Team policy archive. Any revisions to the final document will be recorded on the Document Control Report. To obtain a copy of the archived guidelines, contact should be made with the Maternity team.

13. REFERENCES

NICE (April 2023) Hypertension in pregnancy: diagnosis and management. NICE Collaboration (NICE NG133).

RCOG (2011) Hypertension in pregnancy: the management of hypertensive disorders

during pregnancy. NICE Collaboration (NICE CG107).

RCOG (December 2016) Prevention and Management of Postpartum Haemorrhage (Green-top Guideline No. 52)

NICE (June 2022) Preterm labour and birth. NICE Collaboration (NICE NG25)

NICE (2022) PLGF-based testing to help diagnose suspected preterm pre-eclampsia.

Knight M, Kenyon S, Brocklehurst P, Neilson J, Shakespeare J, Kurinczuk JJ (Eds.) on behalf of MBRRACEUK. Saving Lives, Improving Mothers' Care - Lessons learned to inform future maternity care from the UK and Ireland Confidential Enquiries into Maternal Deaths and Morbidity 2009–12. Oxford: National Perinatal Epidemiology Unit, University of Oxford 2014.

14. PUBLICATION DETAILS

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Directorate/Department responsible for Clinical Guideline	Clinical Support & Specialist Services/Maternity. Ms. Uzma Shaikh Consultant Obstetrics & Gynaecology
Contact details	emad.megaly@nhs.net
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Expiry date	17/07/2027
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Appendix A: BSOTS Hypertension Triage

Royal Devon University Healthcare NHS Foundation Trust

This is not an exhaustive list of presenting symptoms and clinical judgement is required

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Airway compromise
Respiratory rate ≥ 30 or oxygen saturation < 92%
Shock: BP < 80 systolic, HR > 130 bpm
Maternal collapse
Fit
Altered level of consciousness or confusion
Massive haemorrhage
Constant severe pain
Fetal bradycardia
BP > 180 systolic or 115 diastolic x 2 readings

→

1. Transfer immediately to LW or Obs theatre
2. Inform shift leader to inform senior obstetric and anaesthetic medical staff

Shortness of breath or chest pain
Severe headache
Vomiting
Moderate or continuous pain
Moderate bleeding (fresh or old)
Active bleeding
Abnormal MEWS (1x red or 2x yellow values)
BP >160 systolic or >110 diastolic x2 reading
Proteinuria ≥ 3
Fetal heart rate < 110 bpm or >160 bpm
No fetal movements

→

1. Remain in triage room until medical assessment or room on LW available
2. Consider IV access
3. Take blood samples for PET profile +/- G&S/clotting screen (if ALT < 100)
4. Obtain urine sample or urinalysis and urinary protein PCR
5. Complete and categorise CTG (if gestation ≥ 26/40)
6. Inform ST3-7 obstetric medical staff of admission and to attend (re-inform or escalate if no review within 15 minutes)
7. Repeat observations every 15 minutes

Mild pain
Mild bleed (not currently active)
Headache
Altered MEWS (1x yellow value)
BP ≥ 140/90
Proteinuria 1-2+
Normal fetal heart rate
Reduced fetal movements

→

1. Can return to waiting room to await more detailed assessment, unless medical assessment or room available
2. Complete and categorise CTG (if gestation ≥ 26/40)
3. Take blood samples for PET profile
4. Obtain urine sample for urinalysis for PCR
5. Inform ST1-2 obstetric medical staff of admission and to attend (re-inform or escalate if no review within 1 hour)
6. Repeat baseline observations after 1 hour unless altered MEWS, in which case in 30 minutes

Minimal or no pain
No headache
Normal MEWS
BP <140/90
No/trace proteinuria
Normal fetal heart rate
Normal fetal movements

→

1. Can return to waiting room to await more detailed assessment, unless medical assessment or room available
2. Consider completion and categorisation of CTG (if gestation ≥ 26/40)
3. If 3x readings of normal BP (at least 30 minutes apart) and no proteinuria/PCR <30 and not on antihypertensive medication, can be discharged home by MW with appropriate follow-up with INT MW or ANC
4. Inform ST1-2 obstetric medical staff of admission and to attend if not suitable for MW to discharge (re-inform or escalate if no review within 4 hours)

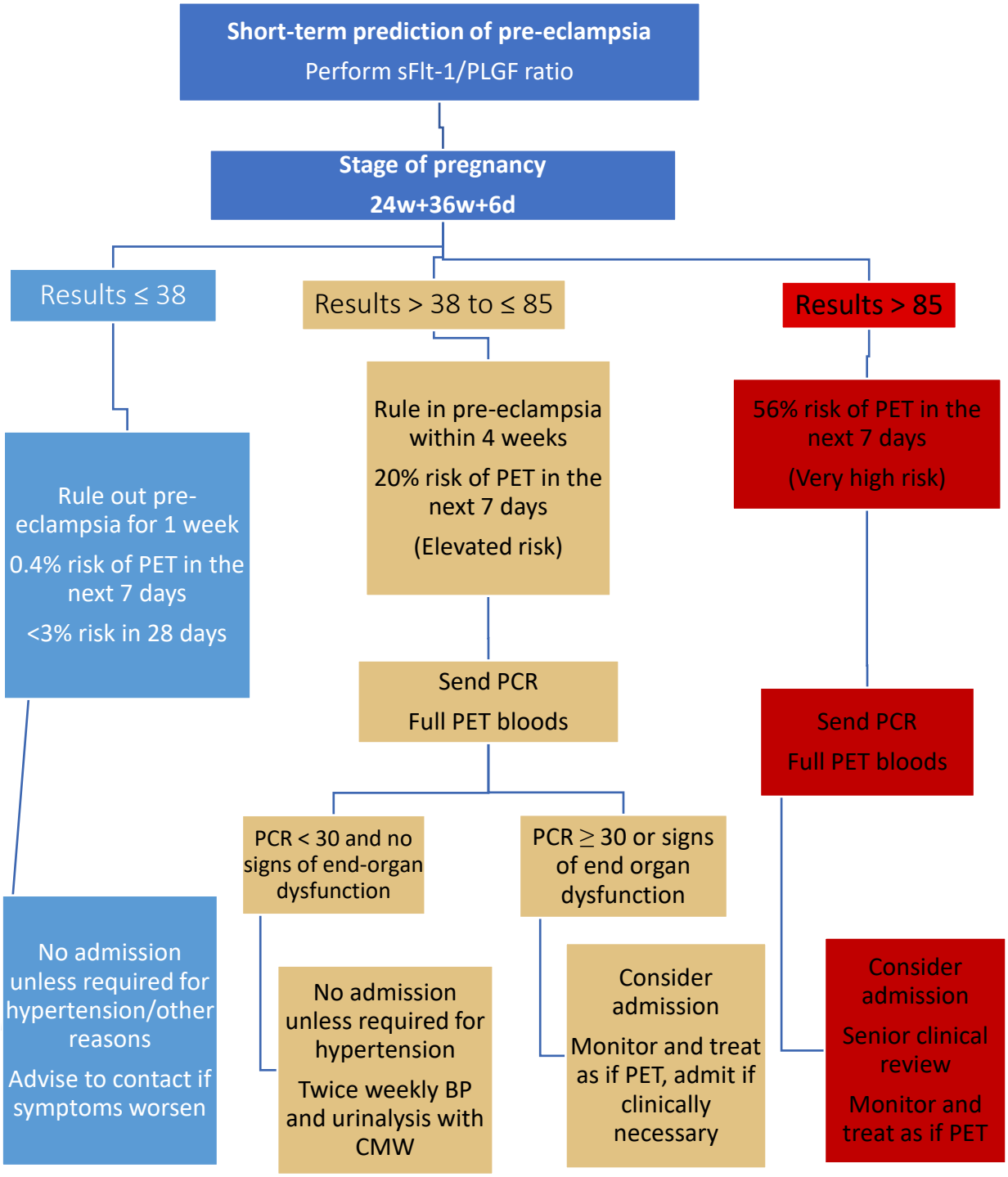
Appendix B: Risk factors for PPH

Discontinue aspirin at 36 weeks gestation if additional risk factors PPH are known:
BMI of 35 kg/m2 or more
Polyhydramnios
Multi-fetal pregnancy
Fetal macrosomia
Placenta praevia
Placenta accreta
Fibroids
Uterine anomalies
Suspected or proven placental abruption
Anaemia
On blood-thinning medication
PPH in a previous pregnancy
History of thromboembolic disease
(If under 16 years of age, decision to start aspirin should be made by a consultant.)

Appendix C:
Clinical Pathway/algorithm-Roche Elecsys immunoassay sFlt-1/PLGF ratio

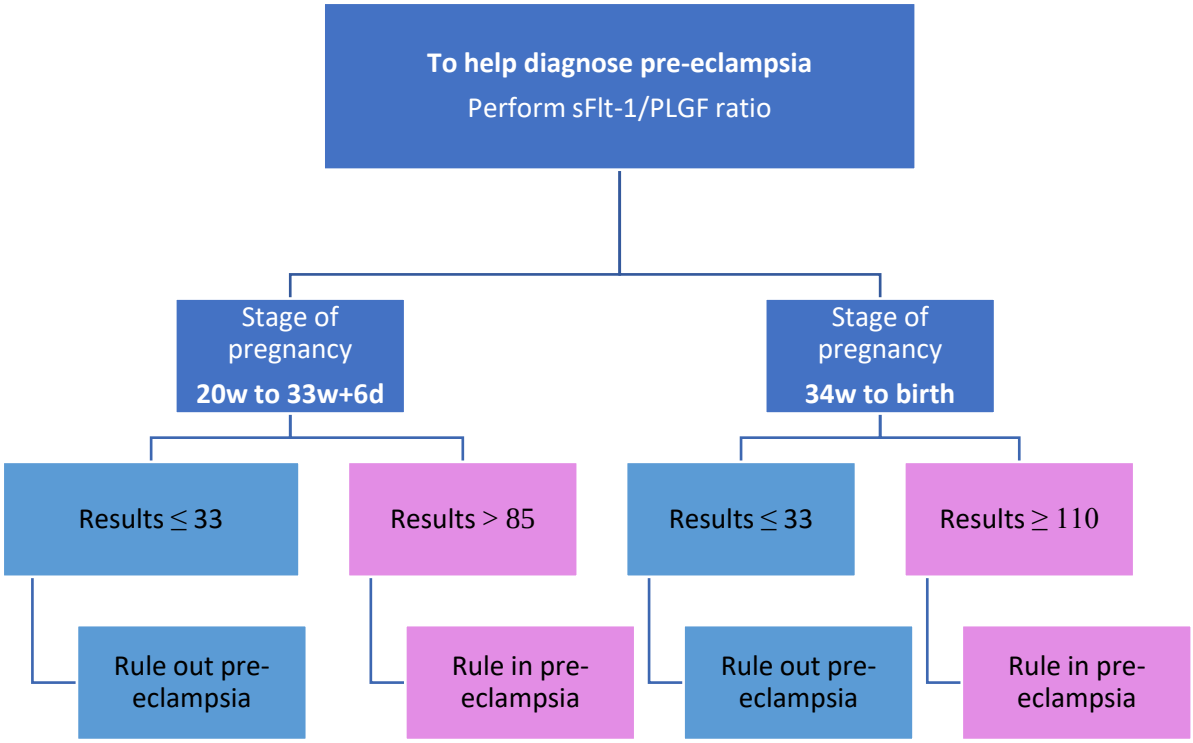
Suspicion of Pre-eclampsia – To help predict

Hypertension/proteinuria/symptoms compatible with PET at **24weeks – 36weeks + 6 days**



Suspicion of Pre-eclampsia – To help diagnose

Hypertension/proteinuria/symptoms compatible with PET at **20weeks – birth**



Recommended cut-offs for the Elecsys immunoassay sFlt-1/PLGF ratio

Intended use	Stage of pregnancy	Decision rule	sFlt-1/PLGF ratio
To help diagnose preeclampsia	Week 20 to week 33 plus 6 days	Rule out cut-off	33
To help diagnose preeclampsia	Week 20 to week 33 plus 6 days	Rule in cut-off	85
To help diagnose preeclampsia	Week 34 to birth	Rule out cut-off	33
To help diagnose preeclampsia	Week 34 to birth	Rule in cut-off	110
Short-term prediction of pre-eclampsia	Week 24 to week 36 plus 6 days	Rule out pre-eclampsia for 1 week	38 or less
Short-term prediction of pre-eclampsia	Week 24 to week 36 plus 6 days	Rule in pre-eclampsia within 4 weeks	Over 38

Appendix D: Anti-hypertensive Therapy Tables

LABETALOL

is the first line treatment at NDDH, oral or intravenous, in the following regimen;
N.B Contraindication- Asthma, pulmonary oedema.

- **Oral Labetalol**
  - 200mg if the Obstetric assessment considers time available for oral intake and therapeutic efficacy, repeat after 30 mins.
  - Maintenance dosage 200mg three times daily (TDS)
- **IV Labetalol**

  BOLUS
  - Give IV 50mg bolus (neat solution) slowly over 5 minutes.
  - Repeat after 5 -10 minutes if necessary.
  - A total of 4 doses (each 50 milligrams) may be given.
  - Do not exceed total maximum cumulative dose of 200 mg.

  MAINTENANCE INFUSION
  - Use Labetalol five milligrams per ml neat solution. Draw up 50 ml neat solution into 50ml syringe and use the BRAUN syringe driver to deliver.
  - Start 20mg/4mls per hour doubling every 30mins to a max of 160mg/32ml per hour.

IV HYDRALAZINE

- BOLUS
  - Dilute 20 mg hydralazine (powder) with 3mls Water for Injection then add 17mls normal saline to total 20mls solution
  - Give hydralazine 5mg (5mls) IV as slow bolus over 15 minutes using the BRAUN syringe driver to deliver at a rate of 20ml per hour. STOP PUMP AFTER 15mins.
  - Do not exceed maximum cumulative dose of 20 mg.
- IV **HYDRALAZINE** MAINTENANCE INFUSION
  - Use 20 mg hydralazine (powder) diluted with 3mls Water for Injection then add 17mls normal saline to total 20mls solution into 20ml syringe and use the BRAUN syringe driver to deliver.
  - Start at 2mg (2mls) per hour and increase by 2mg (2mls) per hour every 30 mins as required to a max of 18mg/hr.
  - NB. Women receiving Hydralazine MUST be warned of the side effects (they can feel particularly uncomfortable and/or painful) including tachycardia, flushing/feeling hot, stuffy nose, tightness in chest, hyperreflexia, abdominal pain, nausea and vomiting, headache which may be severe.
  - STOP Hydralazine if there is significant tachycardia >120bpm and call the SAS Grade or Consultant Obstetrician and senior Obstetric Anaesthetist.
  - Consider using up to 500 ml crystalloid fluid before or at the same time as the first dose of intravenous hydralazine in the antenatal period.

NIFEDIPINE

- Slow Release or Modified Release (SR/MR) **tablets only** NOT Immediate Release (IR) orange capsules.
- Maintenance dosage 10 mg TWICE daily
- Contraindications: aortic stenosis, use within 1 month of myocardial infarction, unstable or acute attacks of angina and acute porphyria.
- Nifedipine increases muscular blockade of magnesium sulphate therefore should NOT be used in conjunction with Magnesium Sulphate.

Medicines and Breastfeeding Interaction Table

No known adverse effects – Assess wellbeing of baby daily for at least 2 days	Dose	Comments
Labetalol	100mg BD, increase to a max of 800mg a day in divided doses	
Nifedipine	10mg BD, increase to max 40mg BD	
Enalapril	5mg OD, increase to 20mg OD if required	Check maternal U+Es one week after starting dose
Captopril	12.5mg BD, increase to 25mg BD	Check maternal U+Es one week after starting dose
Atenolol	25-50mg OD, increase to max 100mg a day in divided doses	
Metoprolol	100mg OD, increase to max 400mg a day in dived doses	
Notes: For all babies whose mothers are taking AHT in the postnatal period asses wellbeing of the baby especially adequacy of breastfeeding at least daily for the first 2 days after birth Insufficient evidence on the safety of: ARB (Angiotensin receptor blockers) Amlodipine ACE other then Enalapri/ Captopril		

Appendix E: Magnesium sulphate (MgSO4) Therapy

Magnesium Sulphate (MgS04) Therapy

Administration of magnesium sulphate to reduce risk of seizure;

Loading dose: 4g $MgSO_4$ as a SLOW BOLUS over 20 minutes

- Draw up 20 mls of 20% $MgSO_4$ solution (4g) into a 50ml syringe
- Place the syringe in a syringe driver and run it at 60ml/hr
- The i.v. infusion will then run over 20 minutes
- In an emergency situation it can also be given as a slow IV push over at least 5 minutes

Maintenance Dose – Magnesium Sulphate **1 gram per Hour**

- Draw up 50ml of 20% $MgSO_4$ (10 g) in a 50 ml syringe
- Give IV using syringe driver at rate of 5ml/hour (1g/hr)

Monitor:

☐ patellar reflexes hourly (or elbow with epidural)
☐ respiratory rate (>10 per min) every 15 mins
☐ O2 Sats >95% every 15 mins
☐ pulse rate every 15 mins
☐ urine output hourly
☐ MgS04 levels if toxicity suspected or persistent oliguria or known impaired renal function

<u>Do not use</u> diazepam, phenytoin or lytic cocktail <u>as an alternative</u> to magnesium sulphate in women with pre-eclampsia.

- Magnesium sulphate is more effective than phenytoin, diazepam and lytic cocktail in preventing the course of pre-eclampsia to eclampsia as well as preventing further eclamptic fits.
- Lytic cocktail is no longer relevant in UK clinical practice.
- Diazepam, phenytoin or other anticonvulsants or anaesthetic agents may be considered by the Consultant Anaesthetist in the ICU setting for women with persistent and unresolved seizures.

Serum Magnesium Levels

- 2-4mmol/L - Therapeutic range
- >4mmol/L - Feeling of warmth, flushing, double vision, slurred speech
- >5mmol/L - Absent reflexes, respiratory depression
- >6mmol/L - Respiratory arrest
- >12mmol/L- Cardiac arrest

Appendix F: Letter to Community on Discharge

This letter can be found on EPIC using the Smartphrase: **.PNHYPERTENSION**

Dear Community Midwife, GP Surgery,

Your patient had hypertension in pregnancy or the postnatal period and has been discharged into the community today.

She delivered on:
Gestational age:

We made the diagnosis of:
☐ Essential hypertension (Pre-existing hypertension)
☐ Pregnancy induced (gestational) hypertension
☐ Pre-eclampsia/ severe pre-eclampsia
☐ HELLP syndrome

Her current antihypertensive medication is:

Medication	Dose	Frequency

The plan of postnatal management:

☐ Essential hypertension
- Check her BP daily for the first two days after birth
- Once between day 3 & 5 after birth or as clinically indicated
- Aim to keep blood pressure lower than 140/90 mmHg
- Review long-term antihypertensive treatment 2 weeks after the birth
- Medical review 6–8 weeks after the birth with the pre-pregnancy care team

☐Pregnancy induced hypertension
- Continue use of antenatal antihypertensive treatment, reduce antihypertensive treatment if her blood pressure falls below 130/80 mmHg
- Check her BP daily for the first two days after birth
- Once between day 3 & 5 after birth or as clinically indicated
- Aim to keep blood pressure lower than 150/100
- If she did not take antihypertensive treatment during pregnancy, start antihypertensive treatment her their blood pressure is higher than 149/99 mmHg
- Should have a medical review 6–8 weeks post birth

☐ Pre-eclampsia
- Continue use of antenatal antihypertensive treatment, reduce antihypertensive treatment if her blood pressure falls below 130/80 mmHg
- Once between day 3 & 5 after birth
- Every 1-2 days up to 2 weeks (if she is on antihypertensives) or until BP is normal (without treatment)
- 6–8 weeks after the birth, the postnatal review must include referral for a specialist assessment if she still needs antihypertensive treatment
- If she has proteinuria (1+ or more) at the postnatal review a further review at 3 months after the birth to assess kidney function and consider offering them a referral for specialist kidney assessment

- Advice use of aspirin 150 mg in next pregnancy from 12 weeks onward

Applies to all cases:

- Refer to **GP** if the patient remains on medication at 2 weeks, if two measurements =/ > 150/100 more than 20 mins apart
- Refer to **hospital** review if symptoms of pre-eclampsia or if BP is >160/100 mmHg.
- She {DID/DID NOT:27244} have significant proteinuria and we {would/not:18918} suggest checking her BP and urine dipstick at 6 weeks postnatal
- If patient remains on medication at 6 weeks postnatal or proteinuria persists, please refer her for appropriate medical review at the hospital
- If antihypertensive treatment is still required following assessment at the postnatal review, specialist assessment should be arranged.

Her most recent blood tests are {NORMAL/ABNORMAL:20511}

Normal Range	**Blood Test**	
44-73	Creatinine	**LFT:** **Renal:** **SFLT**
6-32	ALT	
11-14	Haemoglobin	**FBC:**
150-400	Platelets	

- We would suggest repeating the blood in *** weeks.
- PN follow up

We hope this information is helpful to you for her management in the community setting.